Math 629 Spring 26 **Name: Mengxiang Jiang**

Turn this assginment in on gradescope. All work must be neat and legible. Turn in a "final draft". There should be nothing scratched out and your work should flow generally from left to right and top to bottom.

1. Solve problem 1 of chapter 7 of the Nine Chapters using the method of surplus and deficiency: Several people purchase in common one item. If each person paid 8 coins, the surplus is 3; if each paid 7, the deficiency is 4. How many people were there and what is the price of the item?

   Let n be the number of people and p be the price of the item. We have the following two equations based on the problem statement:

$$8n - p = 3 \quad \text{(surplus when each pays 8 coins)}$$
$$7n - p = -4 \quad \text{(deficiency when each pays 7 coins)}$$

   We use the surplus-deficient formula to find the correct price x that each person should pay:

$$x = \frac{3 * 7 + 4 * 8}{3 + 4} = \frac{21 + 32}{7} = \frac{53}{7}$$

   Now we can substitute x back into one of the equations to find n:

$$8n - xn = 3 \implies 8n - \frac{53}{7}n = 3 \implies \frac{56}{7}n - \frac{53}{7}n = 3 \implies \frac{3}{7}n = 3 \implies n = 7$$

   Finally, we can find the price of the item p:

$$p = 8n - 3 = 8 * 7 - 3 = 56 - 3 = 53$$

   Thus, there were 7 people and the price of the item was 53 coins.

2. Solve the equation $16x^2 + 192x - 1863.2 = 0$. numerically using Qin Jiushao's procedure.

We figure out the first digit of the solution by plugging in some values of x into the equation and checking the sign of the result. If we plug in $x = 7$, we get

$$16 * 7^2 + 192 * 7 - 1863.2 = 784 + 1344 - 1863.2 = 264.8 > 0$$

If we plug in $x = 6$, we get

$$16 * 6^2 + 192 * 6 - 1863.2 = 576 + 1152 - 1863.2 = -135.2 < 0$$

Thus, the first digit of the solution is 6. We now set $x = 6 + y$ and substitute into the equation:

$$\begin{aligned} 16(6+y)^2 + 192(6+y) - 1863.2 &= 0 \\ 16(36 + 12y + y^2) + 192 * 6 + 192y - 1863.2 &= 0 \\ 16y^2 + 384y + 576 + 1152 - 1863.2 &= 0 \\ 16y^2 + 384y - 135.2 &= 0 \end{aligned}$$

Now we repeat the process to find the next digit of the solution. If we plug in $y = 0.3$, we get

$$16 * 0.3^2 + 384 * 0.3 - 135.2 = -18.56 < 0$$

If we plug in $y = 0.4$, we get

$$16 * 0.4^2 + 384 * 0.4 - 135.2 = 20.96 > 0$$

Thus, the next digit of the solution is 3. We now set $y = 0.3 + z$ and substitute into the equation:

$$\begin{aligned} 16(0.3+z)^2 + 384(0.3+z) - 135.2 &= 0 \\ 16(0.09 + 0.6z + z^2) + 115.2 + 384z - 135.2 &= 0 \\ 16z^2 + 393.6z + 1.44 - 20 &= 0 \\ 16z^2 + 393.6z - 18.56 &= 0 \end{aligned}$$

Again, we do the same for z, If we plug in $z = 0.05$, we get

$$16 * 0.05^2 + 393.6 * 0.05 - 18.56 = 1.16 > 0$$

If we plug in $z = 0.04$, we get

$$16 * 0.04^2 + 393.6 * 0.04 - 18.56 = -2.7904 < 0$$

Thus, the next digit of the solution is 4. So far the solution is 6.34. We can continue this process to find more digits of the solution, but we will stop here for now.

3. Solve the system:

$$\begin{aligned} x &= 0 \mod 3 \\ x &= 1 \mod 4. \end{aligned}$$

We can solve this system using the Chinese Remainder Theorem. First, we note that the moduli 3 and 4 are coprime. We find M, the product of the moduli:

$$M = 3 \cdot 4 = 12$$

Let $M_k = \frac{M}{m_k}$ for $k = 1, 2$:

$$\begin{aligned} M_1 &= \frac{12}{3} = 4 \\ M_2 &= \frac{12}{4} = 3 \end{aligned}$$

Let $P_k = M_k \mod m_k$ for $k = 1, 2$:

$$\begin{aligned} P_1 &= 4 \mod 3 = 1 \\ P_2 &= 3 \mod 4 = 3 \end{aligned}$$

Now we need to find the multiplicative inverses x_k such that $P_k x_k \equiv 1 \mod m_k$ for $k = 1, 2$:

$$\begin{aligned} 1 \cdot x_1 &\equiv 1 \mod 3 \implies x_1 = 1 \\ 3 \cdot x_2 &\equiv 1 \mod 4 \implies 3x_2 \equiv 1 \mod 4 \implies x_2 = 3 \end{aligned}$$

Finally, we can substitute everything back into the formula to find x:

$$\begin{aligned} x &= \sum_{k=1}^{2} r_k M_k x_k \mod M \\ &= 0 \cdot 4 \cdot 1 + 1 \cdot 3 \cdot 3 \mod 12 \\ &= 9 \mod 12 \end{aligned}$$

4. Solve the system:

$$\begin{aligned} x &= 0 \mod 11 \\ x &= 0 \mod 5 \\ x &= 0 \mod 7 \\ x &= 4 \mod 9 \\ x &= 6 \mod 8 \end{aligned}$$

We again use the Chinese Remainder Theorem to solve this system. One shortcut we can take is to notice that since x is congruent to 0 mod 11, 5, and 7, we can write x as a multiple of the least common multiple of these moduli:

$$x = 0 \mod \text{lcm}(11, 5, 7) = 0 \mod 385$$

Now we can rewrite the system as:

$$\begin{aligned} x &= 0 \mod 385 \\ x &= 4 \mod 9 \\ x &= 6 \mod 8 \end{aligned}$$

First we find M, the product of the moduli:

$$M = 385 \cdot 9 \cdot 8 = 27720$$

Let $M_k = \frac{M}{m_k}$ for $k = 1, 2, 3$:

$$\begin{aligned} M_1 &= \frac{27720}{385} = 72 \\ M_2 &= \frac{27720}{9} = 3080 \\ M_3 &= \frac{27720}{8} = 3465 \end{aligned}$$

Let $P_k = M_k \mod m_k$ for $k = 1, 2, 3$:

$$\begin{aligned} P_1 &= 72 \mod 385 = 72 \\ P_2 &= 3080 \mod 9 = 2 \\ P_3 &= 3465 \mod 8 = 1 \end{aligned}$$

Now we need to find the multiplicative inverses x_k such that $P_k x_k \equiv 1 \mod m_k$ for $k = 1, 2, 3$:

$$\begin{aligned} 72 \cdot x_1 \equiv 1 \mod 385 &\implies x_1 = 72^{-1} \mod 385 \\ 2 \cdot x_2 \equiv 1 \mod 9 &\implies 2x_2 \equiv 1 \mod 9 \implies x_2 = 5 \\ 1 \cdot x_3 \equiv 1 \mod 8 &\implies x_3 = 1 \end{aligned}$$

Since $r_1 = 0$, we don't actually need to compute x_1. Finally, we can substitute everything back into the formula to find x:

$$\begin{aligned} x &= \sum_{k=1}^{3} r_k M_k x_k \mod M \\ &= 0 \cdot 72 \cdot x_1 + 4 \cdot 3080 \cdot 5 + 6 \cdot 3465 \cdot 1 \mod 27720 \\ &= 82390 \mod 27720 \\ &= 26950 \mod 27720 \end{aligned}$$